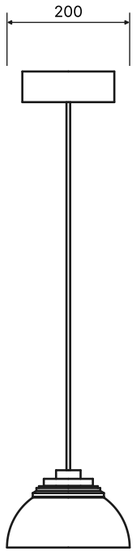
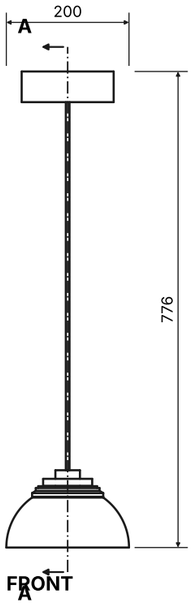
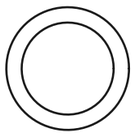
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

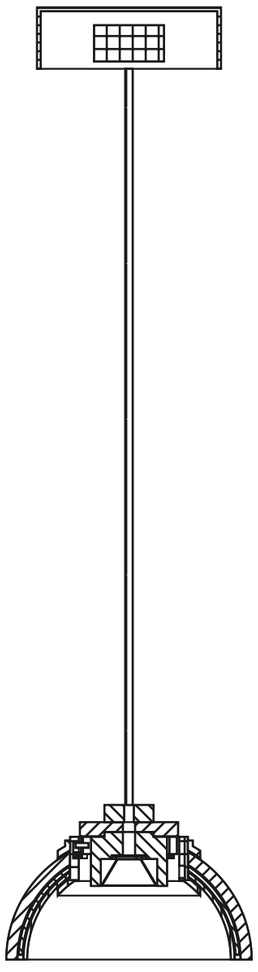
General arrangement

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-10-30 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. Section A-A (sheet 1): the product cut at x 0.0 and seen from the right; cut faces hatched, a different angle for each part.



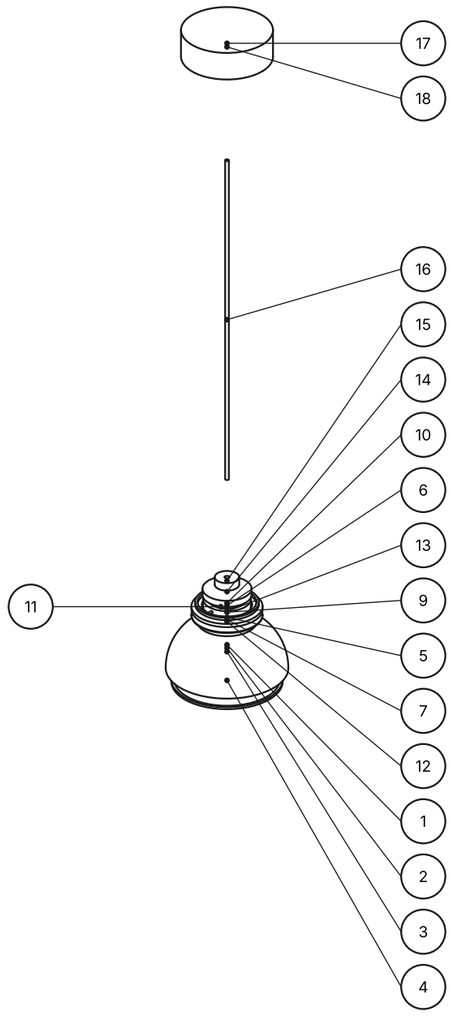
Section A-A

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-10-30 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



Exploded view

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

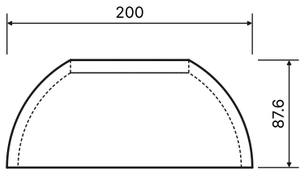
mm | issue 2026-10-30 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

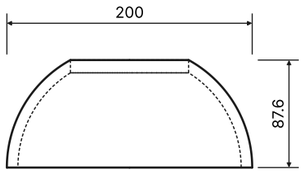
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

1. shell: formed, acrylic-formed; 446 cm3 each; 1 for the set; 1:5

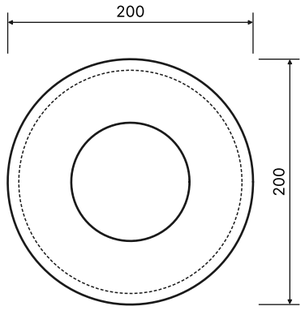
the outside: 10 mm tinted cast sheet drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds (9 at the crown, thinner toward the rim: the former confirms); its outer face never touches the mould, so it keeps the cast sheet's gloss



FRONT



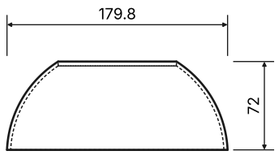
FROM THE RIGHT



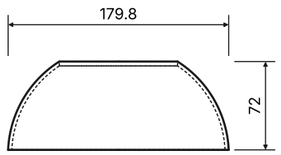
FROM ABOVE

2. liner: formed, acrylic-opal-formed; 117 cm3 each; 1 for the set; 1:5

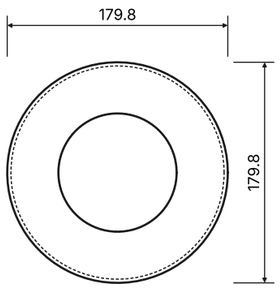
the diffuser: 3 mm volume-diffusing (opal) cast acrylic, light-diffusing grade, about 70 % transmission drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds. Opal scatters through its volume, so it diffuses wherever it touches the tinted layers (a bead-blasted clear liner goes clear where it is bonded or wetted)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

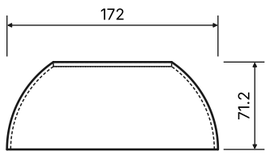
mm | issue 2026-10-30 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

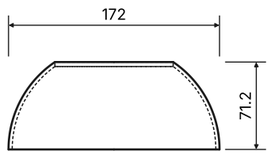
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

3. skin: formed, acrylic-formed; 111 cm3 each; 1 for the set; 1:5

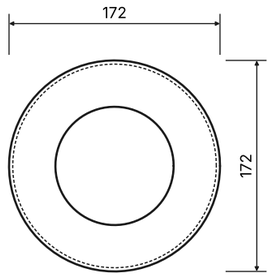
the inside: 3 mm tinted cast sheet formed into a female mould, so its inner (seen) face keeps its gloss and the opal is seen only through tint



FRONT



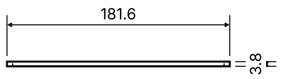
FROM THE RIGHT



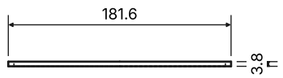
FROM ABOVE

4. rim: lathe, cast-acrylic; 10 cm3 each; 1 for the set; 1:5

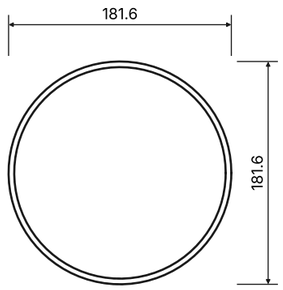
closes the three layers at the mouth, bonded and polished with them; a press-in ePTFE vent in its face keeps the gaps between the layers at room pressure and dry



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

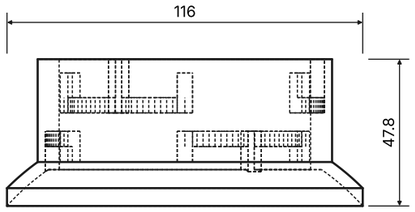
mm | issue 2026-10-30 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

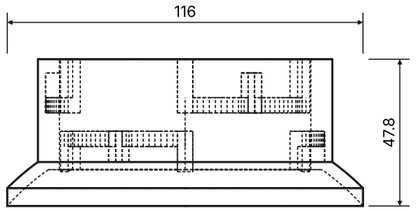
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

5. collar: lathe, aluminium; 84 cm3 each; 1 for the set; 1:2

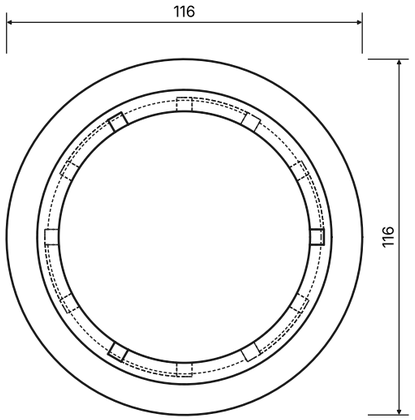
aluminium 6061, black anodised, satin: a tube through the crown with a spherical flange under the skin; three blind L-slots with T-ended notches (the quarter-turn bayonet, 30 deg) work either way up; M96 × 1.5 outside for the clamp ring. It carries the cover mechanically: the bond is a seal only



FRONT



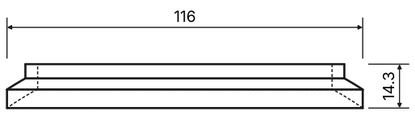
FROM THE RIGHT



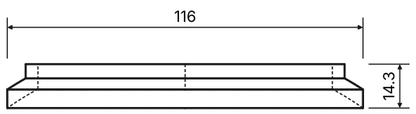
FROM ABOVE

6. clamp: lathe, aluminium; 24 cm3 each; 1 for the set; 1:2

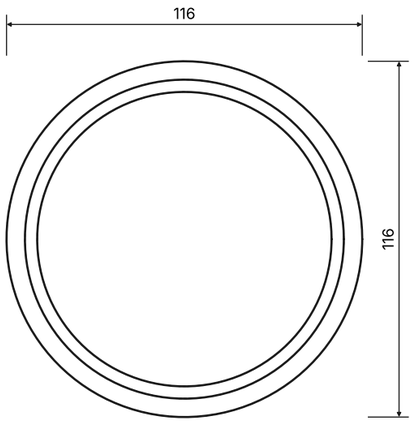
aluminium, black anodised: screwed onto the collar over the crown, its spherical face on a silicone ring, to a torque the shade maker sets (the acrylic is clamped between two silicone rings, never metal on acrylic); a drop of thread-lock



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

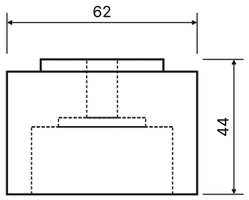
mm | issue 2026-10-30 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

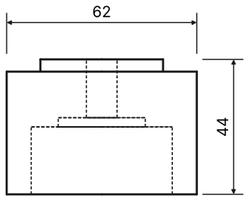
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

9. hub: lathe, aluminium; 86 cm3 each; 1 for the set; 1:2

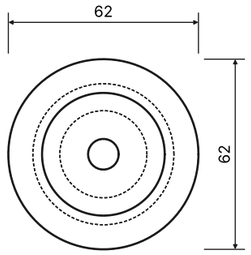
aluminium 6061, black anodised: the core, turned from bar. The COB on its holder at the top of the 46 bore (22 up: the cut-off, about 46 deg from the vertical), the reflector in the bore, the PEEK ring round it, the crown on its neck on a thermal pad; a 10 bore for the wires



FRONT



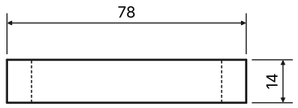
FROM THE RIGHT



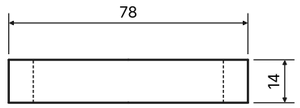
FROM ABOVE

10. ring: lathe, peek; 25 cm3 each; 1 for the set; 1:2

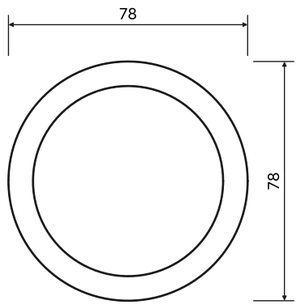
PEEK, unfilled (0.25 W/m K), black: the thermal break. It carries the three pins and the detent, so the cover touches the fixture only through it; 4 to 5 mm of air round the core inside the collar



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

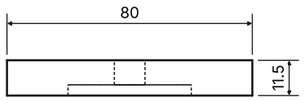
mm | issue 2026-10-30 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

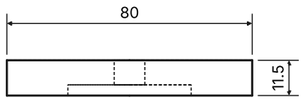
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

14. crown: lathe, aluminium; 53 cm3 each; 1 for the set; 1:2

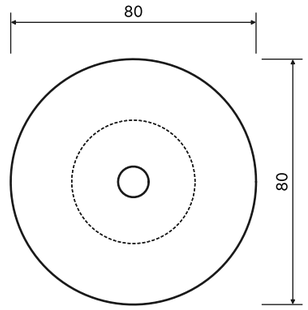
the heat sink (round 41; the fins until then), round 32f's low top: 80 across, 11.5 tall over the collar, turned from 6061 bar, black anodised (emissivity about 0.85: half its loss is radiation), on the core's neck on a thermal pad, 0.5 clear of the PEEK ring and the collar



FRONT



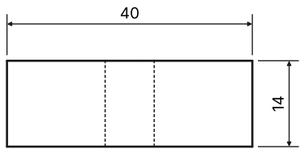
FROM THE RIGHT



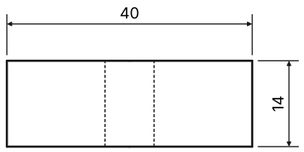
FROM ABOVE

15. gland: lathe, aluminium; 17 cm3 each; 1 for the set; 1:1

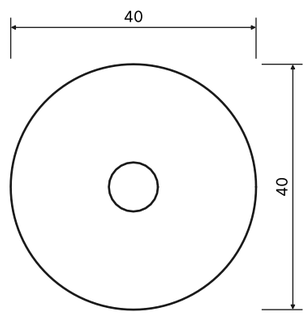
the cap, 40 × 14: aluminium, black anodised and polished (round 32f's was black acrylic: here it runs at the hub's 49 to 52 C), over the listed cord grip (the strain relief, tested with the fixture's weight), on two M3 into the crown; part of the heat sink



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

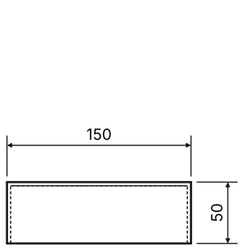
mm | issue 2026-10-30 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

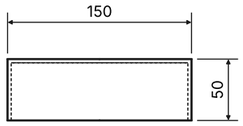
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

17. canopy: lathe, cast-acrylic; 118 cm3 each; 1 for the set; 1:5

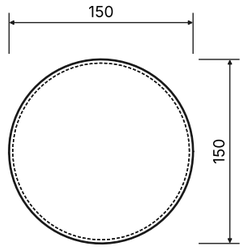
150 × 50, black cast acrylic, polished: over the driver and a steel junction-box strap (the strap carries the load)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-10-30 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	shell	formed	1	Cast acrylic sheet, tinted through, drape-formed over an aluminium mould	446 cm3, 210 × 210 × 98	1,534.41
2	liner	formed	1	Cast acrylic sheet, volume-diffusing (opal), formed over an aluminium mould	117 cm3, 190 × 190 × 82	1,212.73
3	skin	formed	1	Cast acrylic sheet, tinted through, drape-formed over an aluminium mould	111 cm3, 182 × 182 × 81	1,212.30
4	rim	lathe	1	Cast acrylic tube, tinted through	192 × 192 × 14	65.26
5	collar	lathe	1	Aluminium 6061	126 × 126 × 58	67.96
6	clamp	lathe	1	Aluminium 6061	126 × 126 × 24	54.65
7	gaskets	buy	1	Two moulded silicone rings (Shore 40, black, 1 mm, spherical): one between the collar's flange and the skin inside, one between the clamp ring and the shell outside		3.00
8	vent	buy	1	Gore or alike A press-in ePTFE pressure-equalising vent (about 4 across), in the rim ring: it lets the air gaps between the layers breathe and keeps out dust and liquid		1.50
9	hub	lathe	1	Aluminium 6061	72 × 72 × 54	52.00
10	ring	lathe	1	PEEK rod, unfilled	88 × 88 × 24	110.05
11	pins	buy	1	Three stainless dowel pins 4 × 8 (ISO 8734), pressed into the hub: the bayonet's pins		1.50
12	reflector	buy	1	a metal spinner (to quote: Helander Metal Spinning, Lombard IL; Acme Metal Spinning, Minneapolis MN) Round 42: spun to order, no catalogue part fits the bore. A plain conical reflector spun from 0.5 mm spinning-grade high-reflectance		6.00
13	cob	buy	1	Bridgelux Vero SE 10 COB, 2700 K, CRI 90 (BXRC-27G1000-D-73), on an Ideal Chip-Lok holder with a thermal pad, at the top of the hub's bore. DS120: at 350 mA 26.0 V typical, 9.1 W, about 1,130 lm at Tc 25 C and 1,020 at Tc 85 C; at 700 mA (its		12.00
14	crown	lathe	1	Aluminium 6061	90 × 90 × 22	49.35
15	gland	lathe	1	Aluminium 6061	50 × 50 × 24	46.50
16	cord	buy	1	any (UL listed) Black textile-braided pendant cord, 2-conductor 18 AWG, UL listed (SVT-rated, or a listed Class 2 cord: past the driver in the canopy it carries the LED's 26 V DC, not mains), 3 m cut to the drop; a listed cord grip in the cap and		16.00
17	canopy	lathe	1	Cast acrylic tube, tinted through	160 × 160 × 60	106.44
18	driver	buy	1	ERP Power ESS010W-0180-42 (round 42; the 575 lm level): constant current 180 mA, 7.6 W, 24-42 V DC out, 120-277 V AC in; forward-phase (TRIAC) and reverse-phase (ELV) dimming at 120 V AC and 0-10 V, 1-100 %; Class 2; FCC 47 CFR Part 15 Class B		16.00
19	plunger	buy	1	Kipp or Misumi A stainless ball spring plunger M5, light force (about 10 N), threaded into the hub: the bayonet's detent, clicking into a dimple in the collar		4.50
20	listing	buy	1	UL or Intertek (ETL) Listing the pendant to UL 1598 (and the table lamp's battery and charger to UL 153 / IEC 62368): a one-off cost a design, spread over the first run		[PRICE]

Parts list

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

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Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. Hemisphere (the Dome until round 35; Hemisphere North in rounds 35 to 37): the shade hangs mouth down. Its shades swap by hand: lift the shade 5 mm, turn it 30 degrees and lower it off. To fit one, line up the three slots, push up, turn until the plunger clicks, and let go: its weight holds each pin up in its notch, so a knock cannot turn it off. No tools, and the shade is never live (the COB, the hub and the driver stay up).
2. The shade is clamped, not hung on a glue joint: the collar's flange sits under the crown inside, a ring screws down over it outside, the acrylic between two silicone rings. A failed bond cannot drop it.
3. The wall is tint, opal, tint: a 9 mm tinted shell, a 3 mm volume-diffusing opal liner, a 3 mm tinted skin, each formed from cast sheet on an aluminium mould, nested, closed at the mouth by a tinted rim with a vent. Opal scatters through its depth, so it still diffuses where it touches the tint (a bead-blasted clear liner would go clear wherever it was bonded). Nothing white is seen.
4. Light: a Bridgelux Vero SE 10 (2700 K, CRI 90) at the top of a 46 bore lined with a reflector, 22 up: the cut-off about 46 deg from the vertical, and about 85 % of its light out (the black bore passed 52 %). About 575 lm at the COB (the owner, round 36): about 590 at 180 mA (4.4 W) on an ERP Power ESS010W-0180-42 (Class 2, FCC Part 15 Class B), dimmed 100 % to 1 % by a phase-cut (TRIAC or ELV) dimmer or 0-10 V. Not for 1,100 lm: the low top is sized for 575 (1,100 needs the fins of rounds 35 to 39).
5. Heat (round 41): round 32f's low top is the heat sink, no fins: a black anodised aluminium crown 80 across and 11.5 tall over the collar, and an aluminium cap 40 × 14 over the cord grip, the cap's top 126 above the rim. A PEEK ring carries the bayonet's pins, so the hub reaches the shade only through it and air. At 575 lm the hub makes 2.7 W of heat; estimated in a 35 C room: the hub 49 C, the COB's case 50 C, the collar and the acrylic beside it 41 C (hemisphere_thermal.py; to measure, README round 35).
6. Mains stops at the canopy: the driver is there, and the cord carries the LED's 26 V DC (Class 2). The cord is a listed 18 AWG 2-conductor in a black textile braid, its grips tested with the fixture's weight. Listed to UL 1598 with any shade.
7. Hang over a bar at 750 to 900 centres, the rim 700 to 800 above the counter; over a table, one a table, the rim about 650 to 750 above it.

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